Model Comparison on the Cylinder Graph HMM

Claude & C. L.

February 8, 2026

1 Problem Setup

We consider a Hidden Markov Model (HMM) whose hidden-state graph has the topology of a discrete cylinder. The cylinder has *width* $n = 6$ (nodes per level) and *depth* $D = 3$ (number of levels), giving $n \times D = 18$ hidden states in total.

Transition structure. From a node (l, j) at depth l and angular position j the chain can jump to three neighbours:

- **Nearest neighbour (NN):** $(l,\, j{+}1 \bmod n)$ with weight $w \sim \mathrm{Uniform}(0, 1-p)$;
- **Next-nearest neighbour (NNN):** $(l,\, j{+}2 \bmod n)$ with weight $1 - p - w$;
- **Next layer:** $((l{+}1) \bmod D,\, j)$ with fixed probability $p = 0.1$.

The intra-layer weights are drawn once at initialisation (seed 42) and held fixed thereafter; the inter-layer probability p controls how often the chain cycles through the depth levels.

Emission structure. The vocabulary has $|V| = D \times K = 3 \times 16 = 48$ tokens, partitioned into three clusters of 16 tokens each (one cluster per depth level). Each hidden state at depth l emits tokens only from cluster l, with emission probabilities drawn from a symmetric Dirichlet distribution with concentration parameter $\alpha = 0.3$. The small α makes the per-state distributions sparse, so different states at the same depth emit noticeably different token profiles.

Entropy rate. The theoretical (Shannon) entropy rate of the process, estimated via a long Monte-Carlo trajectory, is

$$h \;\approx\; 2.4396 \text{ nats/token.} \tag{1}$$

This serves as the information-theoretic lower bound on the cross-entropy loss achievable by any predictor.

2 Transformer Architectures

We train causal transformer language models (implemented via TransformerLens `HookedTransformer`) to predict the next token in sequences of length $T = 16$ generated by the cylinder HMM.

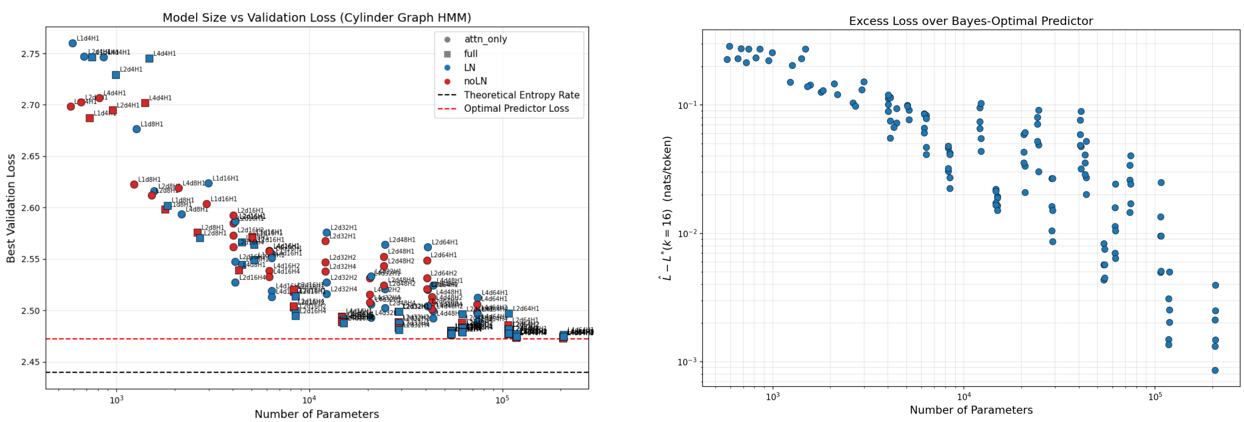
The hyperparameter sweep covers:

Hyperparameter	Values
Number of layers L	2, 4
Embedding dimension d	4, 8, 16, 32, 48, 64
Number of heads H	1, 2, 4
Architecture	attention-only, full (attention + MLP)
Normalisation	none, LayerNorm

All models use ReLU activations, tied input/output embeddings, and $d_{\text{mlp}} = 4d$ for full architectures. Training uses AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 0.01, learning rate 10^{-3}) with batch size 128 for 10 epochs. A total of 135 models were trained on a dataset of $\sim$618k sequences (10^7 tokens).

3 Results: Model Size vs. Validation Loss

Figure 1 plots the best validation loss achieved by each model against its parameter count.



(a) Validation loss vs. parameter count. Circles: attention-only; squares: full (attention + MLP). Blue: LayerNorm; red: no normalisation. Dashed lines mark the theoretical entropy rate $h \approx 2.44$ nats/token and the Bayes-optimal loss $L^*(k{=}16)$.

(b) Excess loss $\hat{L} - L^*(k{=}16)$ on a log–log scale. All architectural distinctions are suppressed to highlight the overall scaling trend with model size.

Figure 1: Model size versus validation loss for 135 transformer configurations on the Cylinder Graph HMM. (a) Absolute validation loss. (b) Excess loss over the Bayes-optimal predictor at context length $k = 16$.

Key observations:

1. **Scaling with model size.** Validation loss decreases roughly log-linearly with parameter count up to $\sim$$10^5$ parameters, after which gains diminish.

2. **Full > attention-only.** At every scale the full transformer (with MLP layers) outperforms its attention-only counterpart, consistent with the MLP layers' role in implementing nonlinear token-mixing needed to approximate posterior belief updates.

3. **Layer norm is approximately neutral.** LayerNorm and no-LayerNorm models achieve similar best losses; the effect is secondary to architecture and scale.

4. **Gap to entropy rate.** The best model (L4, d=64, H=4, full, no-LN; $\sim$206k parameters) reaches a validation loss of ≈ 2.473 nats/token, roughly 0.034 nats above the theoretical entropy rate. This residual gap likely reflects the finite context window ($T = 16$) and architectural limitations in representing exact belief-state inference.

5. **Depth helps.** All top-10 models use $L = 4$ layers, suggesting that additional depth is beneficial for tracking the hidden-state posterior across multiple emission steps.

Log likelihood ratio tests. To evaluate the ability of the model in predicting the next token, we compare between:

1. The true next-token prediction, computing the exact belief state at each step assuming a prior of the stationary distribution. Note that in this case we know the parameters of the HMM.

2. The transformer model's prediction, where the log likelihood of the next token is extracted from the model's logits.

4 Embedding Structure

Figure 2 shows the cosine-similarity matrix and ℓ_2 norms of the learned token-embedding vectors from a representative model. The 3×3 block structure in the similarity matrix directly mirrors the depth-level clustering of the vocabulary: tokens within the same depth cluster are mapped to similar embedding directions, while cross-cluster pairs are near-orthogonal or anti-correlated. This confirms that the model discovers the latent depth structure without supervision.

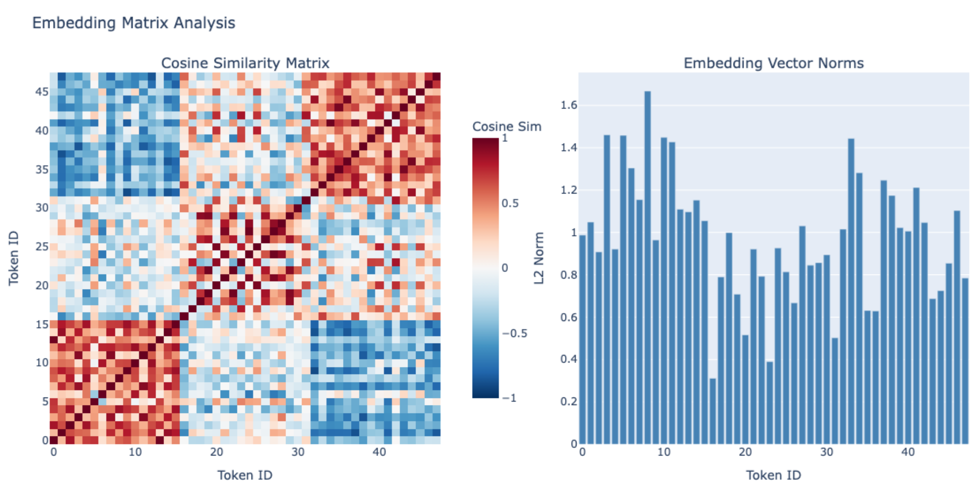


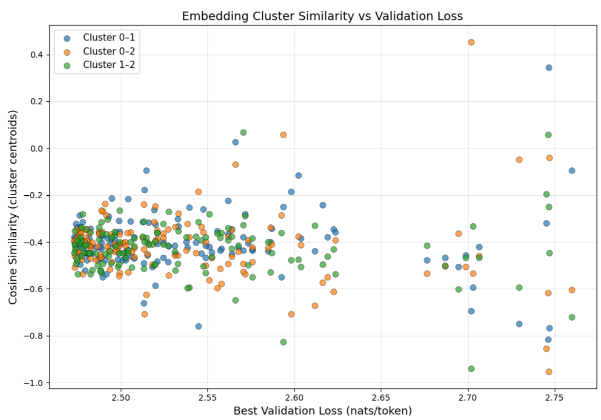
Figure 2: Left: cosine similarity matrix of the $48 \times d$ embedding matrix. The visible 3×3 block structure corresponds to the three depth levels. Right: ℓ_2 norm of each token's embedding vector.

Cluster similarity across architectures. To quantify how consistently the embedding geometry reflects the depth-level structure, we compute the cosine similarity between cluster centroids across all 135 trained models. For each model we extract the embedding matrix $W_E \in \mathbb{R}^{48\times d}$ and compute three centroids

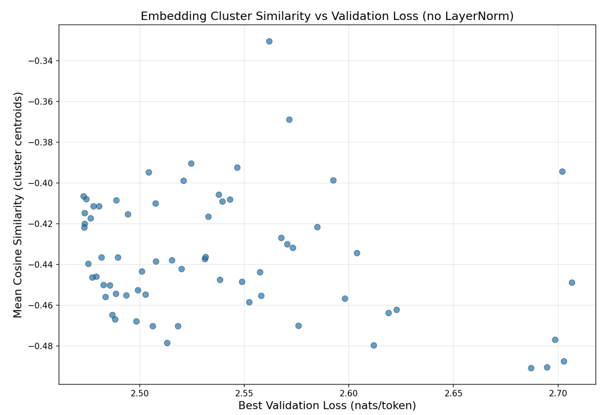
$$\mathbf{c}_l = \frac{1}{16} \sum_{v=16l}^{16l+15} W_E[v,:], \qquad l \in \{0,1,2\}, \tag{2}$$

followed by the three pairwise cosine similarities $\mathrm{sim}(\mathbf{c}_0, \mathbf{c}_1)$, $\mathrm{sim}(\mathbf{c}_0, \mathbf{c}_2)$, and $\mathrm{sim}(\mathbf{c}_1, \mathbf{c}_2)$.

Figure 3 shows these similarities as a function of validation loss. All three pairs are predominantly negative (mean ≈ -0.4), confirming that the learned embeddings separate the three depth clusters into roughly opposing directions in embedding space. Models that achieve low validation loss (left side of the plot) cluster tightly in the range -0.35 to -0.50, while poorly performing models exhibit much higher variance—some cluster pairs even become positively correlated. This suggests that learning to separate the depth clusters in embedding space is necessary for good predictive performance, but the degree of separation saturates once the model is sufficiently expressive.



(a) All three pairwise cosine similarities between cluster centroids, plotted for each of the 135 trained models. Poorly performing models show high variance, with some cluster pairs becoming positively correlated.



(b) Mean of the three pairwise similarities per model, restricted to architectures without LayerNorm (67 models). The trend is flat, with most models in $[-0.49, -0.35]$.

Figure 3: Cosine similarity between depth-cluster centroids of the embedding matrix versus best validation loss. (a) Per-pair similarities for all models. (b) Mean similarity for noLN models only.

5 Bayes-Optimal Loss at Finite Context

The entropy rate h is the minimum achievable loss for a predictor with access to the infinite past. A transformer with context window T, however, can only condition on the most recent T tokens. To disentangle the loss due to *finite context* from the loss due to *limited model capacity*, we compute the Bayes-optimal next-token loss at finite context: the best any predictor can do when restricted to a window of length k.

Setup. Let $\{X_t\}$ be the observed token process generated by the HMM with hidden states $\{S_t\}$, transition–emission matrices $E[v, s, s'] = P(X_t{=}v,\, S_t{=}s' \mid S_{t-1}{=}s)$, and stationary distribution π.

Given a context window $(x_1, \ldots, x_k)$, the Bayes-optimal prediction for the next token is

$$P(X_{k+1}{=}v \mid x_1, \ldots, x_k) \;=\; \sum_{s,s'} \alpha_k(s)\, E[v, s, s'], \tag{3}$$

where α_k is the posterior (belief state) over hidden states after processing the context.

Forward algorithm. The belief state is computed recursively via the forward algorithm. Initialise from the stationary prior $\alpha_0(s) = \pi(s)$. Upon observing token x_t, update:

$$\tilde{\alpha}_t(s') \;=\; \sum_s \alpha_{t-1}(s)\, E[x_t, s, s'], \qquad \alpha_t(s') = \frac{\tilde{\alpha}_t(s')}{\sum_{s''} \tilde{\alpha}_t(s'')}. \tag{4}$$

The normalising constant $Z_t = \sum_{s'} \tilde{\alpha}_t(s')$ equals the predictive probability $P(x_t \mid x_1, \ldots, x_{t-1})$, so the conditional log-likelihood of the context decomposes as $\log P(x_1, \ldots, x_k) = \sum_{t=1}^{k} \log Z_t$.

After k update steps the belief α_k encodes all information that the context provides about the current hidden state. Plugging it into Eq. (3) yields the Bayes-optimal predictive distribution and hence the minimum achievable loss at context length k:

$$L^*(k) \;=\; \mathbb{E}\Big[-\log P\big(X_{k+1} \mid X_1, \ldots, X_k\big)\Big], \tag{5}$$

where the expectation is over sequences drawn from the HMM.

Fixed-context versus infinite-context. For a *fixed* context window, each length-k chunk is treated independently: the belief is reset to π at the start of every window. This is exactly the information available to a transformer with context length k. For the *infinite-context* baseline, the forward algorithm runs over the entire sequence without resetting, so the belief accumulates information from arbitrarily far in the past. The difference

$$\Delta(k) \;=\; L^*_{\text{fixed}}(k) - L^*_{\infty} \tag{6}$$

isolates the cost of the finite context window. As k grows beyond the mixing time of the chain, $\Delta(k) \to 0$.

Decomposing the transformer loss. The cross-entropy loss of a trained transformer $\hat{L}$ can now be decomposed as

$$\underbrace{\hat{L}}_{\text{transformer loss}} \;=\; \underbrace{h}_{\text{entropy rate}} \;+\; \underbrace{\Delta(k)}_{\text{finite-context cost}} \;+\; \underbrace{D_{\text{KL}}}_{\text{model capacity gap}}, \tag{7}$$

where $D_{\text{KL}} = \hat{L} - L^*_{\text{fixed}}(k) \geq 0$ measures the sub-optimality of the transformer relative to the Bayes-optimal predictor at the same context length. Computing $L^*_{\text{fixed}}(k{=}16)$ for the cylinder HMM will reveal how much of the observed $\sim$0.034 nat gap is attributable to finite context versus limited model expressivity.

6 Summary and Open Questions

The cylinder graph HMM provides a controlled testbed for studying how transformers learn to perform approximate Bayesian filtering over a structured latent space. Larger, deeper, full-architecture models approach but do not reach the entropy rate, suggesting fundamental limits tied to the autoregressive architecture and finite context.

Open directions include:

- Characterising the belief-state geometry learned by the residual stream (cf. the fractal structure observed in the belief-state notebooks).
- Ablation studies on context length to quantify the loss attributable to finite T.
- Mechanistic interpretability of attention heads—do specific heads specialise in tracking depth transitions versus intra-layer dynamics?